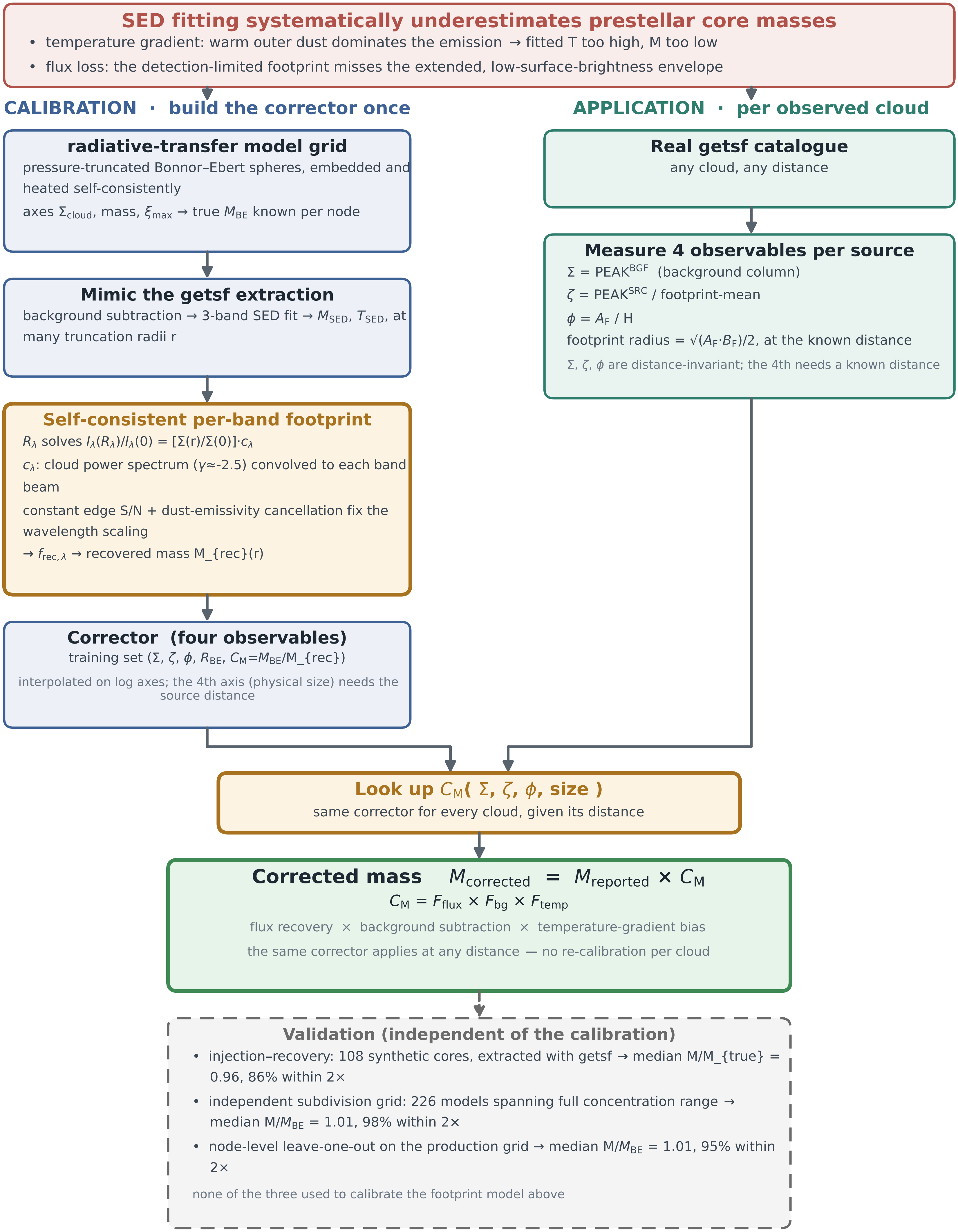
SED fitting systematically underestimates prestellar core masses
• temperature gradient: warm outer dust dominates the emission → fitted T too high, M too low
• flux loss: the detection-limited footprint misses the extended, low-surface-brightness envelope
CALIBRATION · build the corrector once
APPLICATION · per observed cloud
radiative-transfer model grid
pressure-truncated Bonnor–Ebert spheres, embedded and heated self-consistently
axes Σ_{cloud}, mass, $\xi_{\max}$ → true M_{BE} known per node
Real getsf catalogue
any cloud, any distance
Mimic the getsf extraction
background subtraction → 3-band SED fit → M_{SED}, T_{SED}, at many truncation radii r
Measure 4 observables per source
Σ = PEAK$^{\mathrm{BGF}}$ (background column)
ζ = PEAK$^{\mathrm{SRC}}$ / footprint-mean
$\phi = A_{\mathrm{F}}$ / H
footprint radius = $\sqrt{(A_{\mathrm{F}} \cdot B_{\mathrm{F}})}/2$, at the known distance
Σ, ζ, ϕ are distance-invariant; the 4th needs a known distance
Self-consistent per-band footprint
R_λ solves $I_\lambda(R_\lambda)/I_\lambda(0) = [\Sigma(r)/\Sigma(0)] \cdot c_\lambda$
c_λ: cloud power spectrum ($\gamma \approx$-2.5) convolved to each band beam
constant edge S/N + dust-emissivity cancellation fix the wavelength scaling
→ $f_{\mathrm{rec},\lambda}$ → recovered mass M_{rec}(r)
Corrector (four observables)
training set (Σ, ζ, ϕ, R_{BE}, $C_{\mathrm{M}}=M_{\mathrm{BE}}$/M_{rec})
interpolated on log axes; the 4th axis (physical size) needs the source distance
Look up C_{M}(Σ, ζ, ϕ, size)
same corrector for every cloud, given its distance
Corrected mass $M_{\mathrm{corrected}}$ = $M_{\mathrm{reported}} \times C_{\mathrm{M}}$
$C_{\mathrm{M}} = F_{\mathrm{flux}} \times F_{\mathrm{bg}} \times F_{\mathrm{temp}}$
flux recovery × background subtraction × temperature-gradient bias
the same corrector applies at any distance — no re-calibration per cloud
Validation (independent of the calibration)
• injection–recovery: 108 synthetic cores, extracted with getsf → median M/M_{true} = 0.96, 86% within 2×
• independent subdivision grid: 226 models spanning full concentration range → median M/M_{BE} = 1.01, 98% within 2×
• node-level leave-one-out on the production grid → median M/M_{BE} = 1.01, 95% within 2×
none of the three used to calibrate the footprint model above